

WATERMARKING FOR SUMMARIZATION: IMPLEMENTATION, ANALYSIS, AND STRATIFIED WATERMARKING

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ABSTRACT

I study the detection–quality trade-off of soft and hard watermarking for BART-large-CNN summarization on CNN/DailyMail. Increasing the soft watermark bias improves detection but generally degrades summary quality, with further quality losses after detection saturates. Analysis at fixed generation prefixes shows that stronger bias shifts probability away from token sets originally carrying most of the model’s probability mass. Motivated by this observation, I introduce Stratified Watermarking, which favors green tokens separately within a high-probability head and its complementary tail while preserving each region’s total probability. The method requires no additional training and retains the original detector. On 500 evaluation articles, with a threshold calibrated on 200 disjoint unwatermarked summaries, the Stratified configurations selected at common target detection rates of 90%, 95%, and 99% achieve lower GPT-2 perplexity and higher ROUGE and AlignScore than the corresponding soft-watermark configurations. These results suggest that regional probability preservation can reduce the quality cost of watermarking under the evaluated protocol.

1 WATERMARKING EXPERIMENTS AND ANALYSIS

I apply the watermarking algorithm of [Kirchenbauer et al. \(2023\)](#) to BART-large-CNN on CNN/DailyMail. I compare soft and hard watermarks with an unwatermarked baseline and analyze how watermark bias affects detectability and summary quality.

1.1 WATERMARKING METHODS

The following methods are based on [Kirchenbauer et al. \(2023\)](#). At step t , a fixed key and the preceding token determine a pseudorandom green subset $\mathcal{G}_t$ containing a fraction γ of the vocabulary; the remaining tokens form the red subset. Let V denote the vocabulary and $\ell_{t,i}$ the model logit for token i , conditioned on the source article and the summary prefix generated so far.

Soft watermarking (Soft). After temperature scaling, Soft adds a bias δ to every green-token logit:

$$s_{t,i} = \frac{\ell_{t,i}}{T_{\text{temp}}}, \quad q_t(i) = \frac{\exp(s_{t,i} + \delta \mathbf{1}\{i \in \mathcal{G}_t\})}{\sum_j \exp(s_{t,j} + \delta \mathbf{1}\{j \in \mathcal{G}_t\})}. \quad (1)$$

Here T_{temp} is the sampling temperature and δ is the effective bias after scaling. This parameterization is equivalent to a bias $T_{\text{temp}}\delta$ before temperature scaling. The next token is sampled from q_t ; both green and red tokens remain available.

Hard watermarking (Hard). Hard leaves green logits unchanged and sets red logits to $-\infty$. The next token is sampled from the temperature-scaled distribution restricted to the green subset.

1.2 EXPERIMENTAL SETUP

Model. I use BART-large-CNN (Lewis et al., 2020), a checkpoint fine-tuned for summarization, without additional training. Generation uses full-vocabulary multinomial sampling at temperature 0.8, without top-k or top-p filtering. The unmodified model serves as the unwatermarked baseline.

Dataset. I use CNN/DailyMail (Hermann et al., 2015; Nallapati et al., 2016), version 3.0.0, with 200 calibration articles and 500 disjoint evaluation articles from its validation split. All methods use the same evaluation articles and reference summaries.

Detection. The detector reconstructs token colors from each summary and counts distinct adjacent token pairs, excluding transitions involving special tokens. Let T be the number of scored pairs and N_G the number whose second token belongs to the green list determined by the first token. The score is

$$z = \frac{N_G - gT}{\sqrt{Tg(1-g)}}, \quad (2)$$

where $g = \lfloor \gamma|V| \rfloor / |V|$ is the realized green-list fraction. A summary is detected when $z > \tau$. I calibrate τ on 200 unwatermarked summaries at a nominal false positive rate (FPR) of 1%. On the evaluation set, I report the true positive rate (TPR), the fraction of watermarked summaries detected, and the measured FPR, the fraction of unwatermarked summaries detected. ROC-AUC measures discrimination across score thresholds, treating watermarked summaries as positives and unwatermarked summaries as negatives. Outputs with no scored pairs are retained as non-detections.

Quality. I measure reference overlap using ROUGE-1/2/L F1 (Lin, 2004). I compute corpus-level perplexity (PPL) with GPT-2-medium (Radford et al., 2019) on generated summaries, without conditioning on the source article, as a fluency proxy. For factual consistency, I use AlignScore-base (Zha et al., 2023) with `nli_sp` scoring: each summary sentence is matched against source chunks, and its highest support score is averaged across summary sentences. Higher ROUGE and AlignScore and lower PPL are preferred. Implementation details are provided in Appendix A.

1.3 RESULTS AND ANALYSIS

For $\gamma = 0.5$, the calibrated threshold is $\tau = 2.672612$, with a measured evaluation FPR of 0.6% (3/500). Table 4 in Appendix B provides complete numerical results for Unwatermarked, fifteen Soft bias settings, and Hard.

Effect of watermark bias. Figures 1 and 2 show how the bias δ affects detection and summary quality at fixed $\gamma = 0.5$. Increasing the bias strengthens the preference for green tokens, and the observed TPR rises before saturating at high biases. TPR increases from 2.6% at $\delta = 0.5$ to 100% at $\delta = 4.5$.

Figure 2 shows that increasing δ generally lowers ROUGE-L and AlignScore while raising PPL, indicating a deterioration in summary quality across these measures. Even after the observed TPR reaches 100% at $\delta = 4.5$, further increasing the bias continues to degrade summary quality.

Hard provides a contrasting hard-constraint condition in the same figures. It attains 99.8% TPR, with ROUGE-L 14.86, PPL 171.87, and AlignScore 0.252. Soft at $\delta = 4.5$ achieves 100.0% observed TPR with higher ROUGE-L (19.59) and AlignScore (0.391), and lower PPL (73.92). Thus, on this sample, Hard incurs a larger quality cost without improving detection over this Soft configuration.

Effect of green-list fraction. Table 1 shows the effect of varying γ at a fixed bias of $\delta = 2.5$, with a separately calibrated detection threshold for each γ . TPR first increases and then decreases, with the highest observed rates at $\gamma = 0.25$ and 0.5. Larger fractions yield lower PPL and higher ROUGE-L and AlignScore than $\gamma = 0.5$, but reduce detectability. Thus, increasing the green-list fraction does not consistently improve detection.

Summary. These results reveal a recurring trade-off: stronger detection often comes at the cost of summary quality. This raises a central question: can watermarking achieve a desired detection rate while preserving as much generation quality as possible?

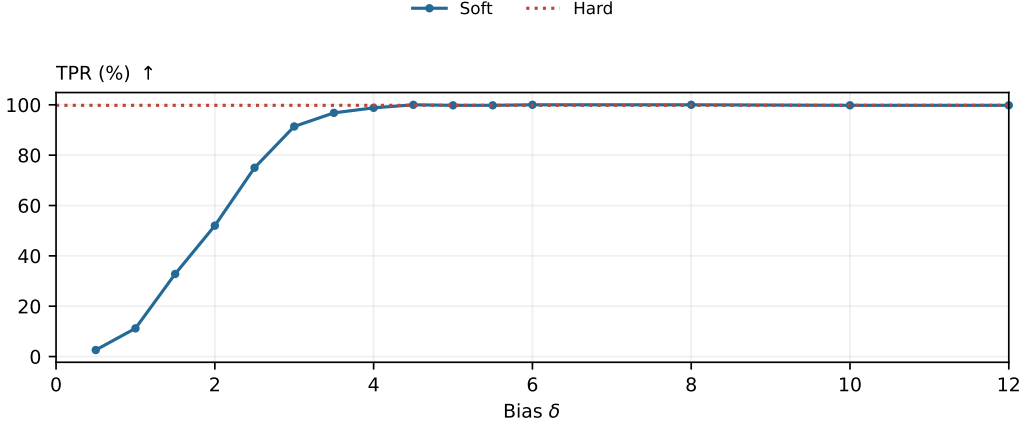


Figure 1: Soft TPR across δ . The horizontal line shows the Hard reference TPR; hard masking has no finite δ .

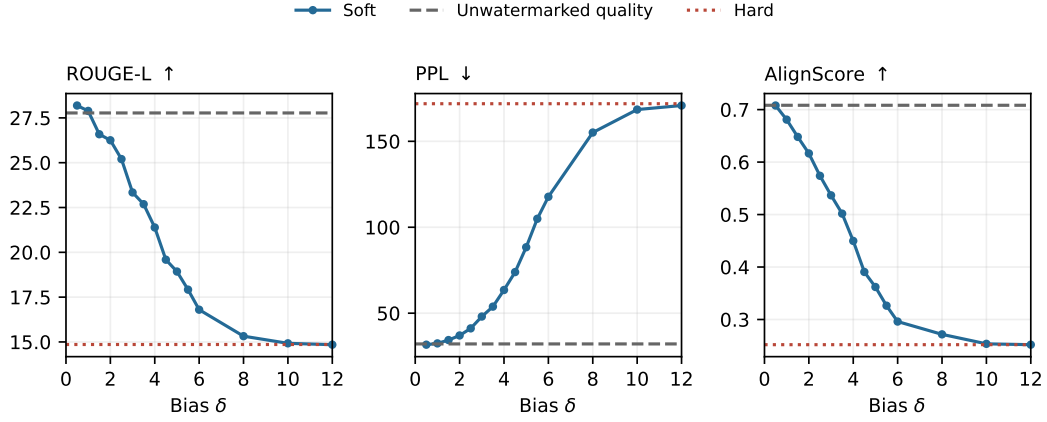


Figure 2: Quality across bias δ at $\gamma = 0.5$. Points cover all fifteen Soft bias settings; lines connect measured settings. Horizontal lines show Unwatermarked ($\delta = 0$) and Hard quality references; Hard has no finite bias parameter.

Table 1: Soft watermark green-fraction sensitivity at bias $\delta = 2.5$. Each fraction uses its own no-watermark-calibrated threshold at the nominal 1% FPR target. R-L denotes ROUGE-L F1 (0–100), and Align denotes AlignScore (0–1).

γ	τ	TPR (%) $\uparrow$	FPR (%) $\downarrow$	AUC $\uparrow$	PPL $\downarrow$	R-L $\uparrow$	Align $\uparrow$
0.1	2.6475	56.6	0.4	0.938	39.05	25.50	0.626
0.25	2.5981	76.2	0.2	0.979	40.65	24.70	0.593
0.5	2.6726	75.0	0.6	0.981	41.16	25.20	0.574
0.75	2.2644	58.2	1.4	0.950	35.66	25.79	0.642
0.9	2.1519	18.6	0.4	0.873	33.51	26.71	0.668

1.4 CHANGES IN PROBABILITY MASS

To investigate potential sources of the quality degradation, I examine how watermarking changes the token probability distribution at individual generation steps. This analysis uses all 500 evaluation articles and 35,795 eligible positions along their unwatermarked summaries. At each position, I compute Soft distributions at the same article and summary prefix, so that differences reflect the watermark transformation rather than different generation histories.

Let p_t denote the temperature-scaled unwatermarked distribution. The original head token set $H_{t,\rho}$ is the smallest set of highest-probability tokens whose cumulative probability reaches ρ before watermarking, including the crossing token. I hold this set fixed and measure the probability mass assigned to it after watermarking:

$$M_{t,\rho}(\delta) = \sum_{i \in H_{t,\rho}} q_t^{(\delta)}(i), \quad \rho \in \{0.90, 0.95, 0.98\}. \quad (3)$$

Here $q_t^{(\delta)}$ is the Soft distribution at bias δ , with $q_t^{(0)} = p_t$. The original mass can exceed ρ because tokens are indivisible. Figure 3 reports probability mass on the original head, averaged across generation positions. The values are actual total probabilities, not ratios to the original mass.

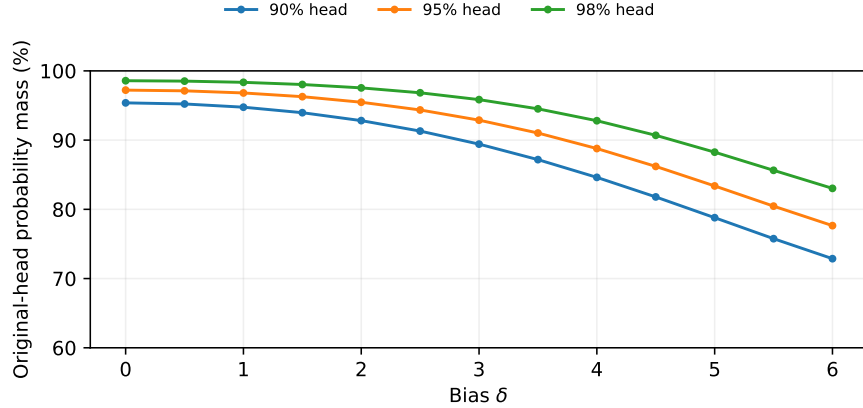


Figure 3: Probability mass on the original head token sets under Soft, averaged across generation positions at identical unwatermarked prefixes. The 90%, 95%, and 98% labels specify cumulative-probability thresholds before watermarking; each token set remains fixed as the bias varies. Initial mass can exceed the threshold because the crossing token is included. Positions within an article are correlated.

Figure 3 shows that, as the bias increases, the token sets originally covering at least 90%, 95%, and 98% of the probability receive substantially less total probability on average after watermarking. At $\delta = 6$, their average total probabilities fall from 95.38% to 72.87%, from 97.22% to 77.65%, and from 98.58% to 83.03%, respectively. The corresponding tail probabilities rise from 4.62% to 27.13%, from 2.78% to 22.35%, and from 1.42% to 16.97%.

Based on these results, I hypothesize that the shift toward the original low-probability tail contributes to the observed quality degradation. By assigning more probability to candidates the model originally considered less likely, watermarking may increase the selection of tokens that fit the source and generated context less well. Such choices can also alter subsequent predictions, allowing local perturbations to accumulate into lower-quality summaries.

2 STRATIFIED WATERMARKING

Motivated by the probability shifts observed in Section 1.4, I introduce Stratified Watermarking, which preserves the total probability of the original head and tail separately while favoring green tokens within each region. This preservation holds at the current generation prefix. The method requires no additional model training.

2.1 PROBABILITY-BASED PARTITIONING

At generation step t , let $p_t(i) = \text{softmax}(\ell_t/T_{\text{temp}})_i$ denote the distribution before watermarking, conditioned on the source article and the current generated prefix. Sort tokens by descending p_t , breaking ties deterministically, and denote their indices by $\pi_t(1), \dots, \pi_t(|V|)$. For a probability-mass threshold $\rho \in (0, 1)$, define

$$k_t = \min \left\{ k : \sum_{j=1}^k p_t(\pi_t(j)) \geq \rho \right\}, \quad H_t = \{\pi_t(1), \dots, \pi_t(k_t)\}, \quad L_t = V \setminus H_t. \quad (4)$$

The head H_t includes the token that crosses the threshold, so its total probability can exceed ρ . The partition is recomputed at every step; ρ controls cumulative probability rather than the number of tokens in the head.

2.2 WITHIN-REGION WATERMARKING

Stratified uses the same green list $\mathcal{G}_t$ as Soft, but normalizes the biased probabilities separately within the head and tail. For each region $B \in \{H_t, L_t\}$, let $P_{t,B} = \sum_{j \in B} p_t(j)$. The sampling distribution is

$$q_t(i) = P_{t,B} \frac{p_t(i) \exp(\delta \mathbf{1}\{i \in \mathcal{G}_t\})}{\sum_{j \in B} p_t(j) \exp(\delta \mathbf{1}\{j \in \mathcal{G}_t\})}, \quad i \in B. \quad (5)$$

Zero-mass regions retain zero probability. The next token is sampled from q_t .

Mass preservation. Summing Equation 5 within a region gives

$$\sum_{i \in B} q_t(i) = P_{t,B}. \quad (6)$$

Thus, the head and tail retain their total probabilities at the current prefix, while individual token probabilities within each region can change. This regional normalization is the key difference from Soft. At $\delta = 0$, the transformation recovers p_t .

Both Soft and Stratified apply the bias after temperature scaling. Stratified is implemented by adding a region-specific log-normalization correction to the biased logits; Appendix A.1 gives the equivalent transformation.

Detection and trade-off. Detection uses the same keyed token colors, z -score, and calibrated threshold as Soft; it requires neither the partition, model logits, nor source article. Preserving regional mass can limit the increase in green-token probability: a region containing only one color remains unchanged, and probability cannot move from a red-only region to green tokens in the other region. Stratified may therefore require a larger bias to reach a given TPR.

3 COMPARATIVE RESULTS

3.1 COMPARISON PROTOCOL

I use the model, articles, quality evaluators, and detection procedure from Section 1. All comparisons use $T_{\text{temp}} = 0.8$ and $\gamma = 0.5$. Soft and Stratified use the same fifteen post-temperature biases: 0.5, 1.0, $\dots$, 6.0, 8, 10, 12. Each configuration is evaluated on all 500 articles. Unwatermarked provides a quality reference, and Hard provides a hard-constraint reference. The common detection threshold is 2.672612, with measured FPR 0.6%.

For the subsequent comparison at a fixed ρ , I select, for each method and target TPR a , the evaluated configuration with the lowest corpus PPL among those meeting the target:

$$\hat{\delta}_m(a) = \underset{\delta \in \mathcal{D}: \widehat{\text{TPR}}_m(\delta) \geq a}{\text{argmin}} \quad \widehat{\text{PPL}}_m(\delta), \quad a \in \{0.90, 0.95, 0.99\}. \quad (7)$$

Here $\mathcal{D}$ is the shared grid and m identifies the method. Ties are resolved by the smaller bias. This rule summarizes the best observed PPL subject to each detection target within the shared grid. Actual TPRs are reported because satisfying a common target does not make TPRs equal. All grid points are retained in the curves and tables.

3.2 EFFECT OF THE HEAD-MASS THRESHOLD

To examine how the partition affects detection, I compare $\rho \in \{0.90, 0.95, 0.98\}$ on the same fifteen-bias grid, using the same 500 articles, temperature, green-list fraction, and calibrated threshold.

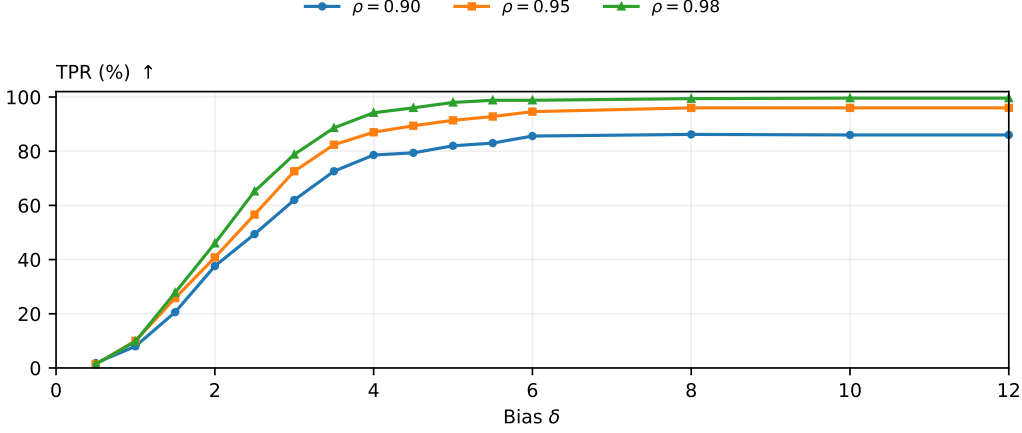


Figure 4: Stratified TPR versus bias for three head-mass thresholds. All curves use temperature 0.8, $\gamma = 0.5$, and the same calibrated threshold (measured Unwatermarked FPR 0.6%). Lines connect evaluated settings; the grid is spaced by 0.5 through 6, followed by 8, 10, and 12.

Figure 4 shows that the three head-mass thresholds lead to different detection plateaus as the bias increases. At $\delta = 12$, TPR reaches 86.0%, 96.0%, and 99.6% for $\rho = 0.90, 0.95$, and 0.98 , respectively. I use $\rho = 0.98$ in the following comparisons because it meets all three target TPRs within the evaluated grid. This choice supports comparison across the target range, without establishing that $\rho = 0.98$ is optimal for summary quality.

3.3 COMPARISON AT TARGET DETECTION RATES

Following Section 3.2, I fix $\rho = 0.98$ for Stratified and select configurations using Equation 7. Complete quality–detection curves over the shared bias grid are provided in Appendix C, with full numerical results in Appendix B.

Table 2: Comparison at target TPRs on 500 articles. Within each method, the lowest-PPL configuration meeting each target is selected from the shared grid. Actual TPRs differ. Stratified uses $\rho = 0.98$; all rows have measured Unwatermarked FPR 0.6%. Parentheses after PPL show the absolute difference from Unwatermarked (32.11). Bold marks the better quality value within each target, using unrounded scores; displayed values may coincide.

Target	Method	δ	TPR (%) $\uparrow$	PPL $\downarrow$	R-1 $\uparrow$	R-2 $\uparrow$	R-L $\uparrow$	Align $\uparrow$
–	Unwatermarked	–	–	32.11	41.41	17.35	27.77	0.71
–	Hard	–	99.8	171.87 (+139.76)	26.63	5.21	14.86	0.25
$\geq 90\%$	Soft	3	91.4	48.05 (+15.93)	37.68	13.14	23.34	0.54
		4	94.2	37.79 (+5.68)	37.88	13.72	23.57	0.57
$\geq 95\%$	Soft	3.5	96.8	53.82 (+21.70)	37.07	12.58	22.69	0.50
		4.5	96.0	38.63 (+6.52)	37.19	12.77	22.74	0.54
$\geq 99\%$	Soft	4.5	100.0	73.92 (+41.81)	33.04	9.36	19.59	0.39
		12	99.6	42.04 (+9.93)	35.11	11.10	20.81	0.48

Table 2 shows that Stratified achieves lower PPL and higher ROUGE-1/2/L and AlignScore than Soft in the configurations selected for all three target detection rates. The gains therefore extend beyond

external language-model perplexity to reference overlap and source consistency as measured by these metrics.

Relative to Unwatermarked, Stratified incurs a smaller PPL increase than Soft at each detection target, including the highest target. Its ROUGE and AlignScore remain below the unwatermarked reference, so the improvement represents a reduction in quality loss. These findings are consistent with the motivation for preserving head and tail probability mass.

4 CONCLUSION AND LIMITATIONS

The results suggest that preserving head and tail probability mass is a useful constraint for reducing the quality cost of watermarking. Stratified improves the measured quality–detection trade-off in the evaluated configurations while retaining the original detector, although it does not fully recover unwatermarked quality.

The study covers one model, one dataset, one key, and one generation-seed protocol; numerical differences do not establish statistical significance. The 200-summary calibration set limits precision at a nominal 1% FPR, and automatic quality metrics do not replace human judgments. Generalization, robustness to editing or adversarial removal, and computational overhead remain to be evaluated.

REFERENCES

- John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A Watermark for Large Language Models. 2023. <https://arxiv.org/abs/2301.10226>.
- Mike Lewis et al. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. ACL, 2020. <https://aclanthology.org/2020.acl-main.703/>.
- Karl Moritz Hermann et al. Teaching Machines to Read and Comprehend. NeurIPS, 2015. <https://arxiv.org/abs/1506.03340>.
- Ramesh Nallapati et al. Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond. CoNLL, 2016. <https://aclanthology.org/K16-1028/>.
- Chin-Yew Lin. ROUGE: A Package for Automatic Evaluation of Summaries. Text Summarization Branches Out, 2004. <https://aclanthology.org/W04-1013/>.
- Alec Radford et al. Language Models are Unsupervised Multitask Learners. OpenAI technical report, 2019.
- Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiyu Hu. AlignScore: Evaluating Factual Consistency with a Unified Alignment Function. ACL, 2023. <https://aclanthology.org/2023.acl-long.634/>.

A EXPERIMENTAL SETTINGS

Table 3 lists the experimental settings. Sample IDs and per-output metadata are stored with the experiment records.

Table 3: Generation, watermark, and evaluation settings.

Item	Setting
Model	facebook/bart-large-cnn; no additional training
Revision	37f520fa929c961707657b28798b30c003dd100b
Data	CNN/DailyMail 3.0.0, official validation split
Samples	200 Unwatermarked calibration articles; 500 evaluation articles
Decoding	Multinomial sampling; do_sample=True, one beam
Temperature	0.8
Candidate filtering	top_k=0, top_p=1.0
Token limits	Input: 1,024; minimum 16 and maximum 192 new tokens
Boundary handling	Decoder start ID 2; forced BOS ID 0 retained; forced final EOS disabled; EOS ID 2; PAD ID 1
Repetition controls	Repetition penalty 1.0; no-repeat n-gram size 0
Inference	FP16, eager attention, batch size 8
Randomness	Seed 1729 + batch starting offset; one seed protocol
Color assignment	$\gamma = 0.5$; key 15485863; lefthash; context width 1
Soft	$\delta = 0.5, 1.0, \dots, 6.0, 8, 10, 12$; bias after temperature scaling
Stratified	$\rho = 0.98$; same post-temperature bias grid; head/tail mass preserved separately
Green-fraction sweep	$\gamma \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$ at $\delta = 2.5$; separately calibrated thresholds
Hard	Red masking; initial forced BOS step exempt; no red fallback
Detector	Re-tokenized text; unique adjacent token pairs; special-token transitions excluded; CUDA color reconstruction
Decision rule	$z > \tau$; calibrated $\tau = 2.672612$ (1% target)
ROUGE	ROUGE-1/2/L F1; stemming; scale 0–100
PPL	GPT-2-medium, FP32; token-weighted corpus NLL; explicit BOS, no scored EOS
AlignScore	AlignScore-base, local nli_sp implementation; saved source articles

The bias is applied after temperature scaling, so δ directly denotes the effective logit tilt. The unwatermarked baseline applies no bias. For Hard, red masking commutes with positive temperature scaling when there is no candidate truncation. For threshold calibration, the cutoff is the $\lceil (n+1)(1-\alpha) \rceil$ -th ordered Unwatermarked score, with $n = 200$, nominal FPR $\alpha = 0.01$, scores ordered from smallest to largest, and a strict greater-than decision rule. The listed threshold refers to $\gamma = 0.5$; each other green fraction has its own threshold. Outputs with no eligible pairs are assigned a detection score of $-\infty$ for threshold and ROC calculations and remain in the sample denominator.

The probability-mass diagnostic uses all 500 evaluation articles and teacher-forced saved Unwatermarked prefixes, FP16 eager model inference, and FP32 probability transformations. It retains minimum-generation constraints and excludes positions whose preceding token is special. This is not bit-identical replay of cached batched generation. Biases are 0, 0.5, 1.0, $\dots$, 6.0, with $\gamma = 0.5$; each head includes its crossing token. The diagnostic is separate from text-only watermark detection and from GPT-2 quality evaluation.

Corpus PPL is $\exp(\sum_s \text{NLL}_s / \sum_s n_s)$, where NLL_s is the total negative log-likelihood and n_s is the number of scored tokens in summary s ; it is not the arithmetic mean of per-summary PPL. AlignScore uses a local inference port: each summary sentence is compared with source chunks, the maximum entailment score is retained, and sentence scores are averaged. Full saved articles supply its evidence, while model inputs are truncated as specified above.

A.1 STRATIFIED LOGIT TRANSFORMATION

For a positive-mass region B , let $s_{t,i} = \ell_{t,i}/T_{\text{temp}}$. An equivalent implementation of Equation 5 adjusts the scaled logits as

$$s'_{t,i} = s_{t,i} + \delta \mathbf{1}\{i \in \mathcal{G}_t\} + \text{LSE}_{j \in B}(s_{t,j}) - \text{LSE}_{j \in B}(s_{t,j} + \delta \mathbf{1}\{j \in \mathcal{G}_t\}), \quad i \in B, \quad (8)$$

where LSE denotes log-sum-exp. The correction preserves each region’s sum of exponentiated logits, so a global softmax yields Equation 5. Zero-mass regions retain zero probability.

B FULL WATERMARK COMPARISON RESULTS

Table 4 provides complete results for both methods on the shared bias grid, together with Unwatermarked and Hard references.

Table 4: Complete shared-grid results on 500 articles at temperature 0.8 and $\gamma = 0.5$. Stratified uses $\rho = 0.98$. Measured Unwatermarked FPR is 0.6%; its TPR and AUC are not applicable. AUC is rounded to two decimals; 1.00 need not indicate perfect separation. R-1/2/L denote ROUGE F1 (0–100), and Align denotes AlignScore (0–1). PPL is reported without baseline differences.

Method	δ	TPR (%) $\uparrow$	AUC $\uparrow$	PPL $\downarrow$	R-1 $\uparrow$	R-2 $\uparrow$	R-L $\uparrow$	Align $\uparrow$
Unwatermarked	–	–	–	32.11	41.41	17.35	27.77	0.71
Hard	–	99.8	1.00	171.87	26.63	5.21	14.86	0.25
Soft	0.5	2.6	0.69	31.68	41.72	17.94	28.19	0.71
	1	11.2	0.81	32.43	41.50	17.60	27.89	0.68
	1.5	32.8	0.90	34.36	40.70	16.37	26.59	0.65
	2	52.0	0.94	37.04	40.57	16.13	26.25	0.62
	2.5	75.0	0.98	41.16	39.59	14.85	25.20	0.57
	3	91.4	0.99	48.05	37.68	13.14	23.34	0.54
	3.5	96.8	1.00	53.82	37.07	12.58	22.69	0.50
	4	98.8	1.00	63.50	35.13	10.99	21.39	0.45
	4.5	100.0	1.00	73.92	33.04	9.36	19.59	0.39
	5	99.8	1.00	88.37	32.02	8.57	18.93	0.36
	5.5	99.8	1.00	104.93	30.90	7.85	17.92	0.33
	6	100.0	1.00	117.73	29.41	6.86	16.80	0.30
Stratified	8	100.0	1.00	155.11	27.17	5.55	15.32	0.27
	10	99.8	1.00	168.47	26.80	5.31	14.92	0.25
	12	99.8	1.00	170.83	26.68	5.21	14.85	0.25
	0.5	1.4	0.68	31.54	41.89	17.92	28.16	0.71
	1	9.8	0.80	31.57	41.85	17.85	28.07	0.70
	1.5	27.8	0.90	32.52	41.23	16.95	27.04	0.68
	2	46.0	0.93	32.29	41.09	16.67	26.70	0.66
	2.5	65.2	0.96	33.72	40.55	15.72	25.85	0.63
	3	78.8	0.98	35.21	39.60	14.79	24.89	0.62
	3.5	88.6	0.99	36.62	38.51	14.06	24.05	0.60
	4	94.2	1.00	37.79	37.88	13.72	23.57	0.57
	4.5	96.0	1.00	38.63	37.19	12.77	22.74	0.54
Stratified	5	98.0	1.00	39.39	36.76	12.40	22.28	0.52
	5.5	98.8	1.00	40.60	36.37	11.86	21.81	0.52
	6	98.8	1.00	41.36	35.88	11.59	21.48	0.51
	8	99.4	1.00	42.35	35.22	11.19	20.90	0.48
	10	99.6	1.00	42.29	35.13	11.11	20.86	0.48
	12	99.6	1.00	42.04	35.11	11.10	20.81	0.48

C ADDITIONAL COMPARISON RESULTS

Figure 5 reports all five quality measures for the shared bias grid. Each point is an evaluated configuration; connecting lines do not imply that intermediate operating points were evaluated. Table 4 in

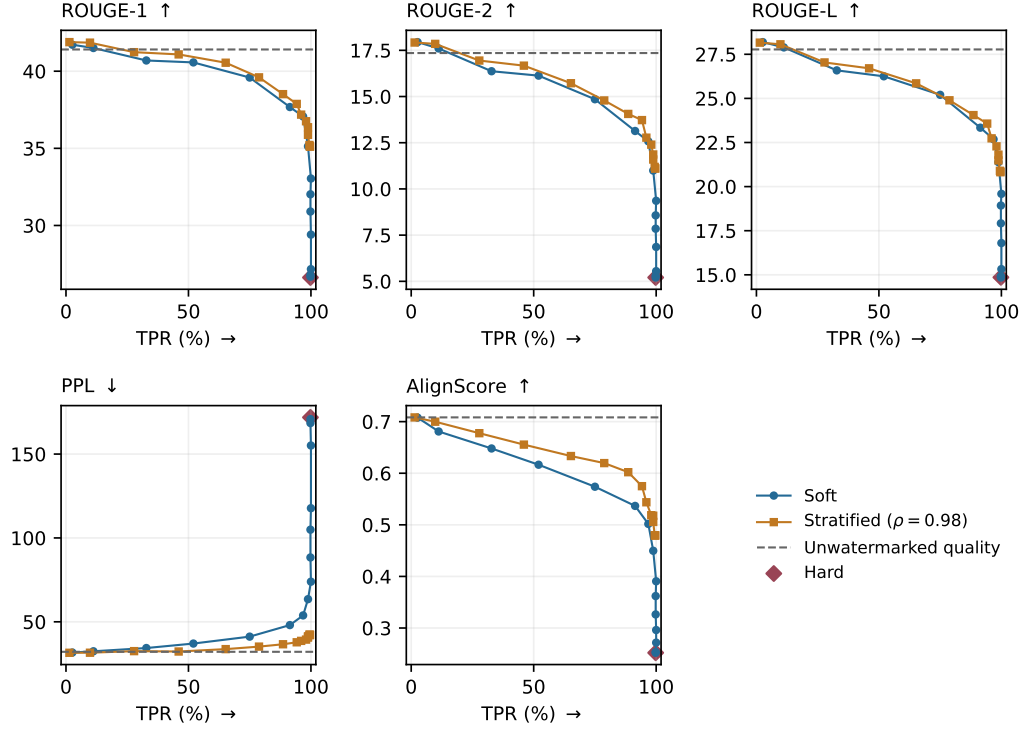


Figure 5: Complete quality–detection curves for the fixed- ρ comparison. Arrows indicate preferred directions: higher TPR, ROUGE, and AlignScore, and lower PPL are better.

Appendix B lists the full shared grid for both methods.